

SMIT VELANI

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Education

Northeastern University, Boston, MA | M.S. Data Science 2026 to 2028 (Expected)
Adani Institute of Infrastructure Engineering (GTU), India | B.E. Information & Communication Technology 2021 to 2025

Experience

Simform Solutions | Data Science Intern Ahmedabad, India | Jun 2025 to Sep 2025
Python, Scikit-Learn, Machine Learning, Data Pipelines, Model Evaluation

- Built end-to-end **machine learning and data pipelines** processing **10K+ daily records**, including data validation, cleaning, exploratory analysis, feature engineering, aggregation, anomaly detection, and reproducible model evaluation.
- Developed predictive models using feature selection, hyperparameter tuning, and cross-validation; evaluated models using **precision, recall, F1, and ROC-AUC** to compare performance across experiments.
- Automated data-quality and model-evaluation workflows and developed KPI-driven analytics to support repeatable analysis and model development.

Unified Mentor Pvt. Ltd. | Data Science Intern Ahmedabad, India | Jan 2025 to Apr 2025
Python, SQL, APIs, NumPy, Pandas, Excel, Power BI, Data Analytics

- Built an end-to-end equities analytics platform integrating **SQL and API data ingestion**, data preparation, quantitative analysis, and interactive reporting.
- Implemented **11 quantitative indicator and return series** using vectorized Python and NumPy, including SMA, EMA, Bollinger Bands, RSI, MACD, ATR, and portfolio risk metrics.
- Delivered a **20-chart analytical dashboard** surfacing market trends, risk-return characteristics, volatility, drawdown, skewness, kurtosis, and other decision-support metrics.

IBM | Data Science Intern Ahmedabad, India | Jun 2024 to Aug 2024
Python, TensorFlow, Keras, OpenCV, Computer Vision, Deep Learning

- Developed a **computer-vision deep learning system** across **16,500+ images and 15 classes** using TensorFlow/Keras, MobileNetV2, OpenCV, and transfer learning.
- Applied a two-phase training and fine-tuning strategy to optimize classification performance, achieving **89% accuracy**.
- Built an inference workflow integrating image preprocessing, feature extraction, trained-model prediction, and a Flask interface for real-time image classification.

Projects

NexusML – Distributed ML Training & Inference Platform | [GitHub](#) *PyTorch, PySpark, Ray, Kubernetes, ONNX, MLflow, FastAPI*

- Built a distributed preprocessing pipeline for **2.4M records and 42 features** across 8 Spark partitions with **12 schema and data-quality gates** and reproducible **70/15/15** train, validation, and test splits.
- Orchestrated **24 PyTorch experiments across 4 Ray workers**, reducing hyperparameter-search time from approximately **96 minutes to 29 minutes**, a roughly **70% reduction**.
- Engineered an asynchronous training service with **5 persisted lifecycle states** and PostgreSQL-backed job metadata, retry handling, failure reasons, configurations, timestamps, and model identifiers.
- Integrated MLflow experiment tracking and model registration, recording **15+ parameters and metrics per run**; automated PyTorch-to-ONNX conversion and validated numerical equivalence across **1,000 samples** with maximum difference below **1e-4**.
- Built ONNX Runtime inference behind FastAPI and benchmarked **10,000+ requests**, reducing p95 latency from **42 ms to 24 ms** and reaching **1,240 predictions/sec** with batch inference.
- Deployed **7 containerized services on Kubernetes** with 2–6 inference replicas, **18 Prometheus metrics**, **12 Grafana panels**, and **64 automated tests** with approximately **85% core-module coverage**.

ActivationLab – Cloud-Native Distributed Platform | [GitHub](#) *Python, Django REST, PostgreSQL, RabbitMQ, Celery, Redis, Docker, Terraform*

- Built a **five-service distributed execution platform** combining REST APIs, PostgreSQL persistence, RabbitMQ/Celery asynchronous workers, Redis caching, Docker, and explicit job lifecycle and failure states.
- Implemented idempotent asynchronous job execution with persisted request state, results, execution attempts, health checks, and structured failure handling.
- Verified the full **API → PostgreSQL → RabbitMQ → Celery → PostgreSQL → API** execution path and runtime idempotency for repeated requests.
- Added automated testing and GitHub Actions CI with **9/9 tests passing**, plus AWS-oriented Terraform infrastructure for container registry provisioning.

- Engineered an end-to-end ML platform for automated data profiling, data-quality validation, target-leakage detection, preprocessing, model comparison, explainability, and drift monitoring.
- Evaluated models on a **284K-record fraud dataset**, achieving **0.981 ROC-AUC and 0.848 PR-AUC**; implemented cost-aware model selection to account for highly imbalanced outcomes.
- Integrated MLflow experiment tracking, SHAP explainability, PSI drift detection, FastAPI model serving, Docker, automated regression testing, and CI/CD.

CausalLens – Causal Inference & Experimentation Platform | [GitHub](#)

Python, Statsmodels, SciPy, Scikit-Learn, Streamlit

- Built an experimentation platform implementing **8 causal and experimental estimators**, including CUPED, difference-in-differences, propensity score matching, AIPW, Bayesian A/B testing, uplift modeling, and sequential inference.
- Reduced estimation bias from **1.82 to -0.05** with AIPW across **30 validation runs**; implemented estimator-specific diagnostics to test assumptions rather than relying only on final point estimates.
- Validated externally on the LaLonde NSW benchmark, recovering an estimated treatment effect of **\$1,637** against **\$1,794** experimental truth, with **45 automated pytest tests**.

AgentAudit – AI Agent Evaluation Framework | [GitHub](#)

Python, LangGraph, LangChain, Scikit-Learn, Pydantic, AI Agents

- Built a ReAct-style **AI agent evaluation framework** combining tool calling, structured outputs, trace logging, a hand-written evaluation set, automated scoring, regression testing, and adversarial testing.
- Validated LLM-based evaluation against human scoring, achieving **0.774 Cohen's kappa** judge-human agreement.
- Reduced unsafe responses from **86% to 57%** with layered guardrails and validation; **23/23 pytest tests passed** under CI.

Global Economic Intelligence Dashboard | [GitHub](#)

Python, Flask, Scikit-Learn, ARIMA, D3.js, WebSockets

- Built a real-time economic intelligence platform covering **190+ countries**, integrating World Bank economic data, interactive visualization, machine learning, forecasting, anomaly detection, and live news sentiment.
- Implemented ARIMA GDP forecasting, K-Means economic clustering, Isolation Forest anomaly detection, VADER sentiment analysis, and WebSocket-based data refresh with interactive D3.js visualizations.

Technical Skills

Programming & Systems: Python, C++, Java, Go, SQL, R, JavaScript, TypeScript, C#, Bash, Linux, Object-Oriented Programming, Data Structures & Algorithms

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, Hugging Face Transformers, BERT, Deep Learning, Computer Vision, Transfer Learning, Feature Engineering, Model Training, Hyperparameter Tuning, Cross-Validation, Model Evaluation

ML Systems & Distributed Computing: Apache Spark, PySpark, Ray, Databricks, Distributed Training, Parallel Processing, Data Pipelines, MLflow, Model Registry, ONNX, ONNX Runtime, Model Serving, Batch Inference, Inference Optimization

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, FAISS, LLMs, Llama 3.x, OpenAI APIs, AI Agents, Tool Calling, Structured Outputs, Pydantic, Guardrails, LLM Evaluation, Regression Testing

Backend, Cloud & Distributed Systems: FastAPI, Flask, Django REST, REST APIs, Microservices, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Celery, Docker, Kubernetes, AWS, Terraform, Asynchronous Processing, Horizontal Scaling

Data Science & Analytics: Pandas, NumPy, SciPy, Statsmodels, Causal Inference, A/B Testing, Statistical Analysis, Time-Series Forecasting, Anomaly Detection, Excel, Power BI, D3.js

ML Reliability & Observability: Prometheus, Grafana, SHAP, Structured Logging, System Monitoring, Latency & Throughput Benchmarking, Data Validation, Model Monitoring, Failure Handling, Health & Readiness Probes

Testing & DevOps: Git, GitHub, GitHub Actions, CI/CD, pytest, Unit Testing, Integration Testing, End-to-End Testing, Docker Builds, Automated Model Validation